

2.11 stddef.h

The stddef header defines several standard definitions. Many of these definitions also appear in other headers.

Macros:

```
NULL
offsetof();
```

Variables:

```
typedef ptrdiff_t
typedef size_t
typedef wchar_t
```

2.11.1 Variables and Definitions

`ptrdiff_t` is the result of subtracting two pointers.

`size_t` is the unsigned integer result of the `sizeof` keyword.

`wchar_t` is an integer type of the size of a wide character constant.

`NULL` is the value of a null pointer constant.

`offsetof(type, member-designator)`

This results in a constant integer of type `size_t` which is the offset in bytes of a structure member from the beginning of the structure. The member is given by *member-designator*, and the name of the structure is given in *type*.

Example:

```
#include<stddef.h>
#include<stdio.h>

int main(void)
{
    struct user{
        char name[50];
        char alias[50];
        int level;
    };

    printf("level is the %d byte in the user structure.\n"),
        offsetof(struct user, level);
}
```

The output should be:

level is the 100 byte in the user structure.